

# Dive Buddy Alert Band

Stay in touch. Know where to look.



A reef draws one diver's attention. Their buddy turns to share something else. Dive Buddy Alert Band brings that small but important connection to the wrist: a private call, a clear acknowledgement, and a proposed view of where the other diver is.

## A private call. A useful view.

Rev F combines a paired attention request with proposed horizontal bearing, relative depth and a shared-position map. Dedicated Dive and Boat views keep elapsed time, motion and the reference position in context.

**Direction and map views require the proposed acoustic positioning reference, plus depth and heading sensing on the bands. All interface readings shown here are simulated. DBA remains an aid to buddy procedures, never a safety guarantee.**

# A shared dive. Two different points of view.

The moment before visual communication.

Underwater, a hand signal depends on being seen. A buddy can be nearby yet outside the other diver's field of view, above the reef or a few metres deeper. Getting their attention is the first step. Understanding where to look is the next.

## One press starts the conversation.

Pair the bands before entering the water. A deliberate press requests a vibration on the paired band. The receiving diver presses to acknowledge. The caller sees that human response separately from the band's automatic delivery receipt.

## Distance gains context.

The proposed navigation view combines separation with horizontal bearing and a clear shallower-or-deeper cue. A second view places both divers on a horizontal grid. Dive information and the boat-associated reference each have a dedicated screen, so every number has a purpose.

The intended use is a routine buddy connection during a shared dive. The band does not replace visual contact, established separation procedures or a dive computer.

Illustrative scenario. No customer testimonial, founder incident, sales result or demonstrated underwater performance is claimed.

# Buddy bearing and relative depth

A horizontal direction with a separate vertical relationship.



11.5 metres is the separation. The arrow and wrist-relative angle locate the buddy horizontally. "2.0 m shallower" supplies the depth relationship. North-up mode expresses the bearing as a compass direction and angle.

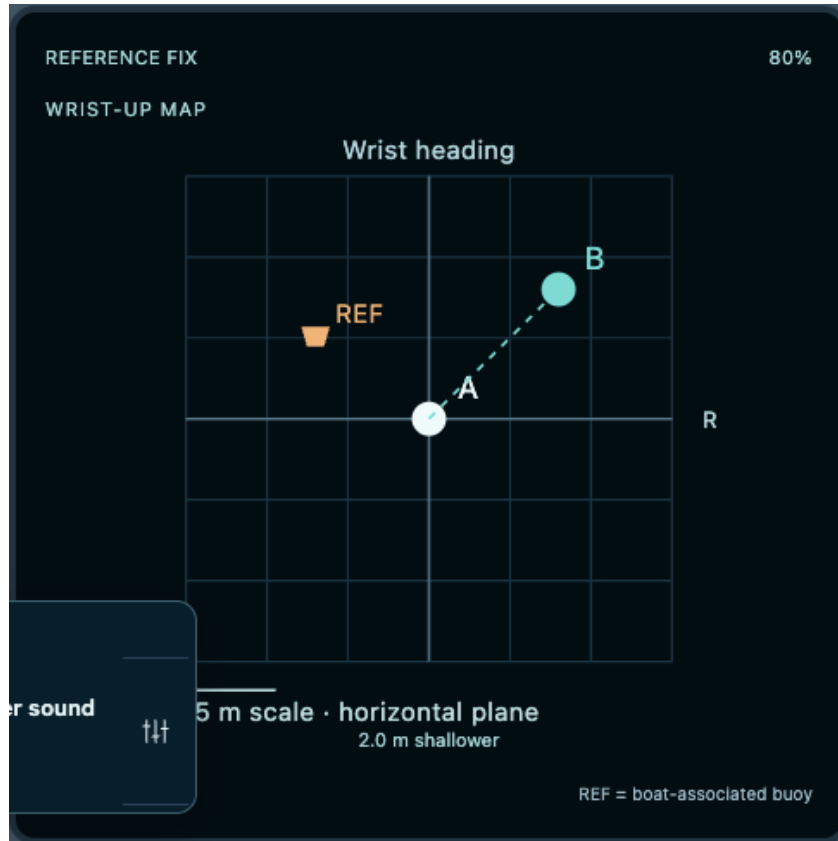
## An arrow needs more than range

Direct ranging still uses one wrist transducer. The proposed boat-associated acoustic reference supplies shared horizontal positions; pressure and heading sensing are added to each band. Heading is the wrist reference, which may differ from swimming direction.

Same horizontal position clears the directional arrow. A shallower/deeper cue describes where a buddy is, not an instruction to ascend or descend. All example readings are synthetic.

# Both divers on one map

A horizontal plane, with depth kept explicit.



The map centres the wearer and places the paired diver and boat-associated reference on a scaled grid. North-up uses east and north axes; wrist-up rotates the view around the wearer's heading.

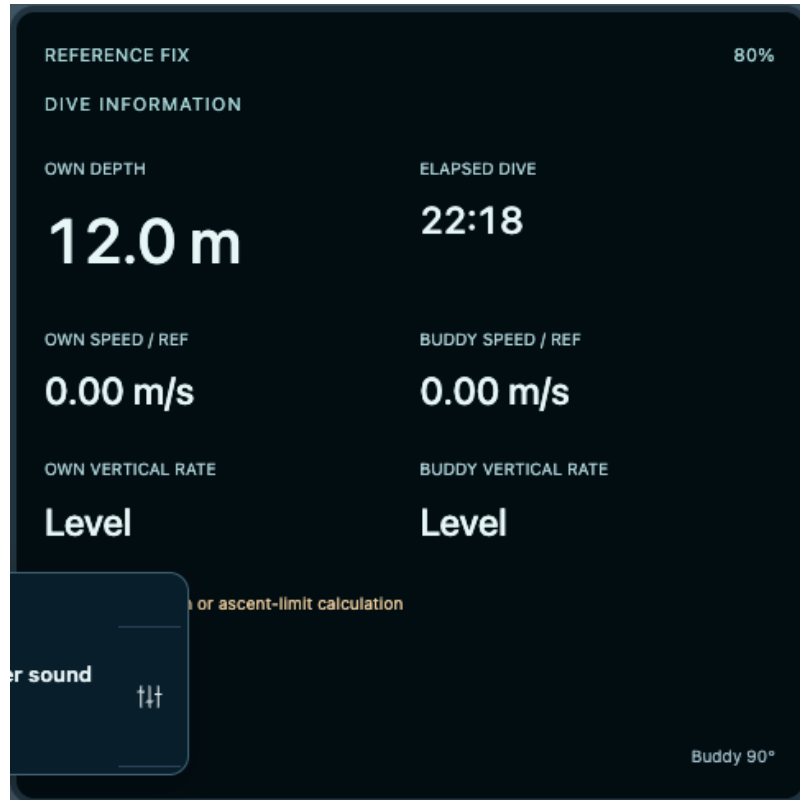
## A map point is not a route

The grid represents relative horizontal positions. It does not contain reef geometry, hazards or a safe path. The below-map depth cue remains separate from the horizontal projection. A current peer data link and valid fixes for both divers are required to show the buddy marker.

Position updates can be paused in the demonstration. After three seconds, stale position markers and arrows clear. That illustrative age limit does not establish a real sensing update rate or accuracy.

# Time, depth and movement

Each measure answers a different question.



Dive information shows own depth, elapsed dive time, both reference-relative horizontal speeds, both vertical rates and wrist headings. The moving and ascending scenarios make those distinctions visible.

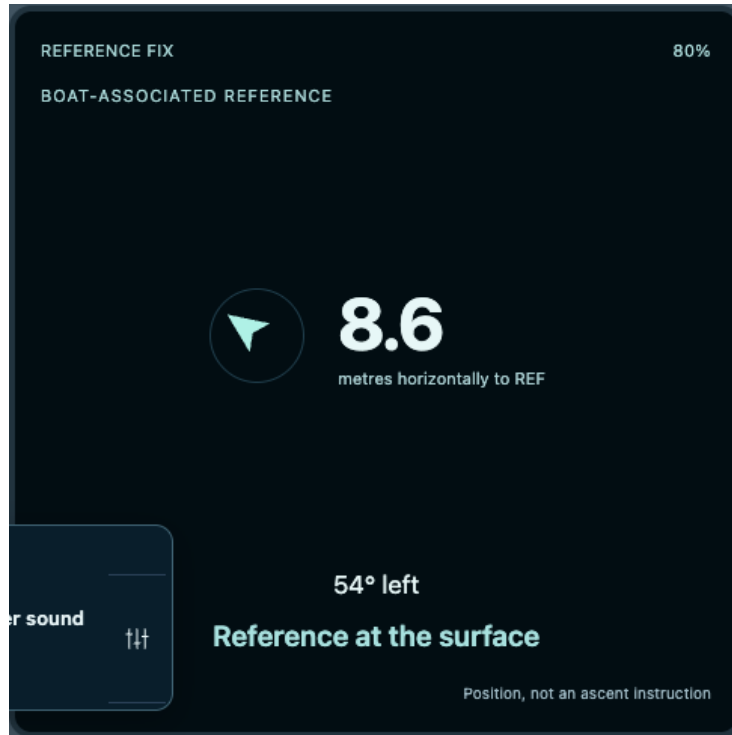
Horizontal speed is relative to the reference, not speed through water. Vertical rate is derived from depth change over time. Neither is a decompression calculation. An ascent-rate readout cannot establish prevention of decompression sickness.

Local depth and time remain when the acoustic reference is unavailable. Exchanged buddy depth and motion require a current peer data link. Reference-derived horizontal speeds clear when their fixes expire.

The example clock starts at 22:18 and can play a 20-second movement sequence. The interface is a simulation, not a diving instrument.

# A reference back at the boat

Proposed equipment and the architecture behind navigation.



The Boat view shows horizontal distance and bearing to the associated reference. A reference mounted or deployed near the boat must have a defined relationship to it. A drifting or repositioned reference changes the meaning of that position.

## Reference array to shared positions to band displays

A spaced acoustic receiver array and orientation reference would support localization. Band depth and wrist heading complement those positions. This is a proposed architecture: array dimensions, aperture, synchronization, calibration, motion compensation, accuracy and coverage remain undefined.

The trade-off is extra equipment, deployment and power. The reference is represented as a system diagram; this package does not supply invented hardware CAD or claim compatibility with an existing manufacturer's system.

**Distance is horizontal to the reference, not a safe ascent instruction. Loss of the reference clears the boat arrow and position.**

# One physical button

Proposed interaction and independent data availability.

## Pair before the dive

Surface pairing establishes the buddy identity. The browser demonstration starts paired. Pair allocation and nearby-pair isolation require firmware definition.

## Short press: select a view

At idle, short presses cycle Buddy, Map, Dive and Boat. Web tabs expose those choices directly. The physical interface remains one top button, not a touchscreen.

## Hold one second: call

A hold requests three pulses on the paired band. The caller first sees Calling, then device receipt. The receiving diver short presses to acknowledge; only that action establishes a human response.

## Cancel and return

A hold cancels an active call. A short press returns from an acknowledged call to the selected view. When the peer link fails, remote cancellation remains unconfirmed.

## Data failures

Peer loss clears buddy range and exchanged buddy data. Reference loss or stale positioning clears arrows, map points and reference speeds. Current direct range, own depth and dive time can remain available independently.

## Low battery

The demo lowers A to a synthetic 12% and suspends A's positioning. This is proposed UI behavior; it does not establish a real low-battery threshold or reserve.

# Adjust the fit. Keep the connection.

Mechanical adjustment and three nominal strap layouts.

A split silicone strap wraps around the wrist or suit cuff. Its overlapping tail adjusts through a proposed acetal buckle, with a silicone keeper securing the loose end. The sealed module remains removable when the strap needs replacing.



Compact, Regular and Cuff are three modeled layouts, with nominal internal reference diameters of 50, 62 and 78 mm. Their circular reference circumferences are about 157, 195 and 245 mm. These dimensions describe CAD layouts; they are not established wrist-size ranges.

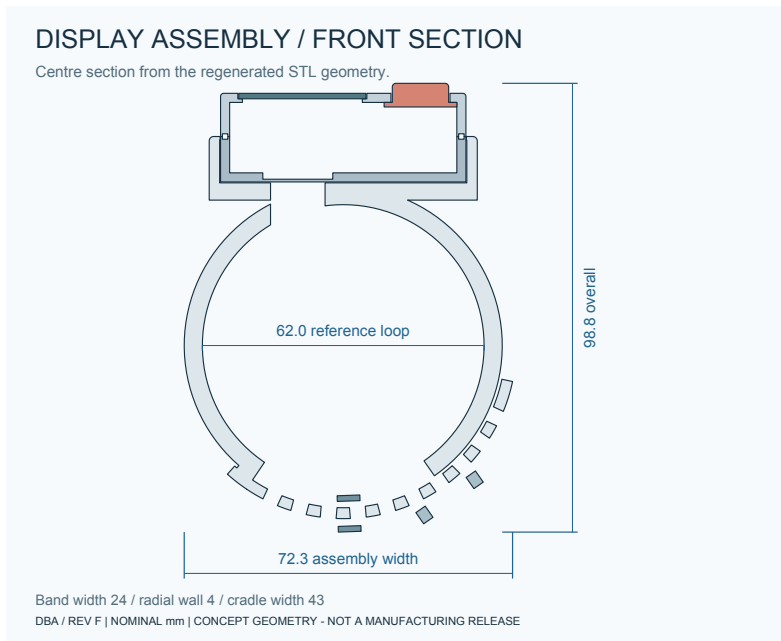
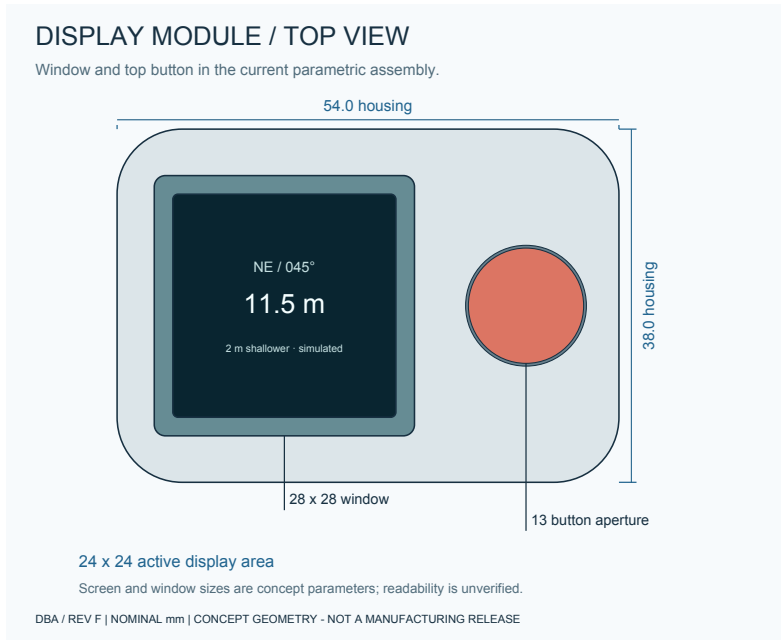
An overlapping silicone tail has ten modeled adjustment holes, with a buckle envelope and tail keeper. The main strap is 24 mm wide and 4 mm thick; the tail is 22 mm wide and 2.4 mm thick.

Corrosion resistance is a design requirement. The proposed exterior combines silicone, an acetal buckle with nonmetal retention, and a polymer housing and window. No exposed steel pin or charging contacts are specified. Material grades, seal durability, buckle strength and the usable fit range remain unvalidated.



# The current assembly

Rev F nominal dimensions, derived from the parametric source.



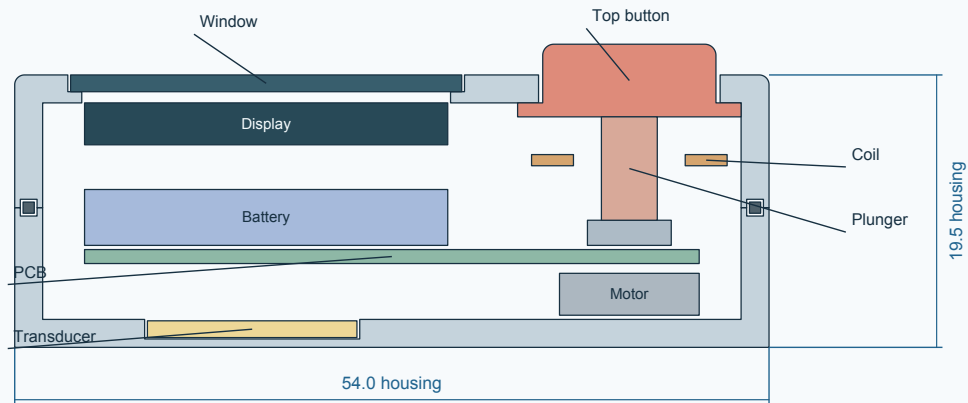
54 x 38 x 19.5 mm housing, excluding 2.2 mm button protrusion. Regular assembly height: 98.8 mm. 18 component envelopes. Compact and Cuff variants move the same module onto different reference strap layouts.

# Inside the sealed module

Display, pressure sensing, heading sensing and charging.

## DISPLAY MODULE / CENTRE SECTION

Current STL section. Color identifies component envelopes.



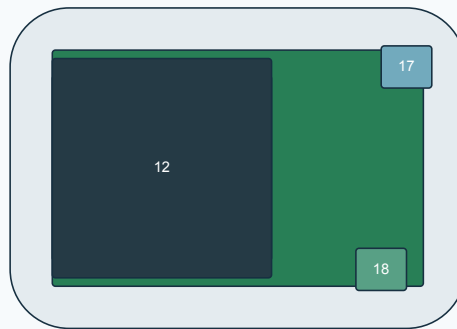
Coil is beneath the button side of the cap; the plunger passes through its centre opening.

Nominal clearances are modeled. Wiring, supports, sealing and charging efficiency remain undefined.

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

## INTERNAL LAYOUT / CAP REMOVED

Plan projection of selected component envelopes; depths differ.



05 PCB · 07 Battery · 12 Display · 17 Pressure sensor · 18 Heading / inertial sensor

Pressure coupling, wiring, mounting and actual component selections remain undefined.

The external acoustic reference is a system proposal; no fabricated array CAD is supplied.

DBA / REV F | NOMINAL mm | CONCEPT GEOMETRY - NOT A MANUFACTURING RELEASE

Display body: 26 x 26 x 3 mm. Window: 28 x 28 x 1.2 mm. Active area: 24 x 24 mm. Sensor parts are envelopes, not selected devices. Pressure coupling, wiring, supports, seals, optical stack and manufacturing tolerances remain undefined.

# Engineering specification

Register 1-5 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

## Product scope / Rev F concept

Paired attention, acknowledgement and reference-assisted buddy navigation. An aid to buddy procedures, never a safety guarantee.

**Engineering notes:** No demonstrated underwater performance, safe-route guidance, reunion or decompression protection.

**Discipline:** Systems / product

## Form and service / Design decision

Adjustable silicone strap and keeper; proposed acetal buckle with nonmetal retention; sealed removable module and one silicone top button.

**Engineering notes:** No exposed steel pin is specified. Buckle latch, attachment, retention and service details remain undefined. Strap replacement does not make the sealed module user-openable.

**Discipline:** Mechanical / industrial design

## Charging interface / Design decision

Inductive charging through the sealed cap; no band charging port or exposed charging contacts.

**Engineering notes:** Receiver coil is beside the display and around the button plunger. Dock alignment, ferrite, thermal limits and charging efficiency require definition.

**Discipline:** Electronics / mechanical

## Direct acoustic range / Baseline architecture

Approximately 65 kHz two-way time-of-flight with one transducer on each wrist.

**Engineering notes:** A direct scalar range cannot produce bearing. Transducer, directivity, drive levels, receiver sensitivity and protocol remain unselected.

**Discipline:** Underwater acoustics

## Range and timing / Historical targets

20 m direct-range target. Rev A assumed approximately  $\pm 0.3$  m range performance, 1 Hz exchange and fixed 2 ms reply delay.

**Engineering notes:** Unmeasured targets; none establishes reference-position accuracy, coverage or update rate. Timing precision alone does not establish accuracy.

**Discipline:** Acoustics / firmware

# Engineering specification

Register 6-10 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

## Operating envelope / Unverified

Rev A proposed 50 m operating depth. The current concept has no pressure rating, validated runtime or assembled mass.

**Engineering notes:** Immersion, temperature, salinity and pressure convention require definition. Earlier four-dive runtime and 28 g mass do not carry over to Rev F.

**Discipline:** Systems / mechanical

## Pairing / Proposed behavior

Surface-only pairing identifies the buddy before the dive. The demonstration begins paired.

**Engineering notes:** Pairing gesture and address allocation need definition; nearby-pair isolation, authentication and reacquisition are unresolved.

**Discipline:** Firmware

## Call and acknowledgement / Proposed behavior

Hold the top button for 1 s to call. The receiver feels three pulses; a short press acknowledges. Device receipt and human acknowledgement are separate states.

**Engineering notes:** Web call buttons simulate the hold. Retry limits, delayed packets, simultaneous calls and timeout behavior need protocol definition.

**Discipline:** Firmware / interaction

## View selection and cancel / Proposed behavior

At idle, short presses cycle Buddy, Map, Dive and Boat. Incoming calls interrupt the view. A hold cancels an active call; an acknowledged call returns to the view with a short press.

**Engineering notes:** Web tabs expose view selection directly. Gesture arbitration, glove actuation and remotely confirmed cancellation remain unresolved.

**Discipline:** Firmware / interaction

## Haptic vocabulary / Revised proposal

Three pulses request attention. Optional demonstration distance pulses become more frequent at shorter range.

**Engineering notes:** Demo cadence is illustrative, not the old Rev A release waveform. Amplitude, suit transmission, hysteresis and distinguishing alerts require definition; close does not mean reunited.

**Discipline:** Interaction / firmware

# Engineering specification

Register 11-15 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

## Missing and stale data / Proposed behavior

Loss of peer range clears buddy range and exchanged data. Loss or age-out of the reference fix clears arrows, map positions and boat position. Independent depth and time remain.

**Engineering notes:** Demo uses a 3 s position-age limit and a synthetic 20 m cutoff per acoustic leg. These illustrate failures; actual quality thresholds, latency, uncertainty and outlier rejection remain unestablished.

**Discipline:** Systems / firmware

## Acoustic position quality / Open design work

A spaced reference receiver array and orientation reference would estimate shared horizontal positions from acoustic exchanges.

**Engineering notes:** Array geometry, clock synchronization, calibration, motion compensation, near-vertical geometry and multipath rejection need design. No locator hardware or measured accuracy is supplied.

**Discipline:** Underwater acoustics / estimation

## Power budget / Pending estimate

Display, depth sensing, heading sensing and reference-assisted exchanges add loads to the original band concept.

**Engineering notes:** 150 mAh / 3.7 V remains a historical reference, not selected capacity. New band and reference power budgets, reserve and charging time are required.

**Discipline:** Electronics

## Battery behavior / Proposed behavior

Local battery remains visible. The demo reduces A to 12% and suspends A's positioning; peer range and local dive data remain.

**Engineering notes:** 12% is a synthetic UI state, not a validated threshold or reserve. Cell protection, cold-water behavior and battery life are unestablished.

**Discipline:** Electronics / firmware

## Housing, seals and corrosion resistance / Design requirement / nominal geometry

Saltwater corrosion resistance is required. Proposed polymer housing and nonmetal window; internal electronics isolated from seawater. Housing: 54 x 38 x 19.5 mm, plus 2.2 mm button protrusion; 2 mm shell wall.

**Engineering notes:** Resin grades, seal interfaces, pressure-sensor water coupling and acoustic membrane require design. Saltwater durability, UV resistance and pressure performance are unvalidated; no corrosion-proof rating is claimed.

**Discipline:** Mechanical / acoustics

# Engineering specification

Register 16-20 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

## Internal packaging / Rev F nominal layout

18 envelopes include display, window, plunger, adjustment buckle, keeper, pressure sensor and heading/inertial sensor.

**Engineering notes:** Nominal non-intersection does not establish wiring, PCB routing, sensor pressure port, supports, actual components or fastener design. Reference array remains a system diagram, without fabricated CAD detail.

**Discipline:** Mechanical / electronics

## Strap and button / Rev F nominal geometry

Compact / Regular / Cuff reference diameters: 50 / 62 / 78 mm. Main strap: 24 mm wide, 4 mm thick. Overlap tail: 22 mm wide, 2.4 mm thick, ten 3.5 mm holes.

**Engineering notes:** 157 / 195 / 245 mm circular circumferences are reference layouts, not wearer ranges. Proposed UV-stabilized acetal buckle grade, creep, latch strength, silicone formulation, safe extension and suit compression remain undefined.

**Discipline:** Industrial design / mechanical

## Drawing authority / Source-derived dimensions

Regular assembly is 72.3 mm wide and 98.8 mm high. All three size variants derive from dba\_cad.py.

**Engineering notes:** Rev F drawings and meshes represent current geometry. Prior revisions are historical. Dimensions describe envelopes, not manufacturing readiness.

**Discipline:** Mechanical

## Commercial case / Hypothesis

Two bands, inductive dock and a boat-associated acoustic positioning reference. Buddy pairs and dive operators are the proposed audience.

**Engineering notes:** Reference deployment, shared use, compatibility, demand and operational value require customer development. Published diver commentary is not a DBA customer validation study.

**Discipline:** Product / commercial

## Economics / Historical estimate only

Rev A display-free design: approximately \$86 per pair at 5,000 pairs,  $\pm 30\%$ ; \$179-199 retail target. No current Rev F system price.

**Engineering notes:** Original estimate excludes tooling, engineering, freight and duty. It does not include the navigation display, added sensors or acoustic reference.

**Discipline:** Commercial / manufacturing

# Engineering specification

Register 21-24 of 24. Proposed behavior, nominal dimensions and unresolved implementation.

## Project status / Presentation and CAD Rev F

Concept by Rafael. Current files contain four interactive navigation views and three parametric adjustable strap layouts.

**Engineering notes:** Model consistency and browser behavior are verified separately from physical performance. No parts procurement or physical test program is commissioned.

**Discipline:** Project lead

## Display and readability / Rev F envelope

Display body 26 x 26 x 3 mm; window 28 x 28 x 1.2 mm; active area 24 x 24 mm.

**Engineering notes:** Enlarged screen studies do not prove legibility at actual scale. Display technology, resolution, contrast, viewing angle and optical stack remain unselected.

**Discipline:** Interaction / electronics / mechanical

## Bearing, depth and orientation / Proposed sensing

Reference-derived horizontal bearing; pressure-derived relative depth; heading sensing supplies wrist orientation and heading difference.

**Engineering notes:** Wrist heading is not necessarily swimming direction. Same horizontal position yields no directional arrow. A compass heading alone does not locate a buddy.

**Discipline:** Systems / acoustics / sensors

## Dive time and motion / Proposed information

Dive view shows elapsed time, own depth, both reference-relative horizontal speeds and both vertical rates. Boat view shows horizontal distance and bearing to the associated reference.

**Engineering notes:** Reference-relative speed is not speed through water. Vertical rate is not a decompression calculation or bends prevention. Boat position assumes a defined relation between boat and reference; neither arrow nor depth cue is an ascent instruction.

**Discipline:** Firmware / estimation / interaction

# What published diver feedback adds

Published observations informed the revision; no direct interviews were conducted.

## Attention is useful; direction adds utility

The Scuba Diver Life Buddy-Watcher review discusses the value of an addressed underwater alert and the lack of directional information. DBA Rev F responds with a proposed positioning architecture, not an assertion that a simple alert band already provides bearing.

<https://scubadiverlife.com/gear/gear-review-buddy-watcher/>

## An alert still depends on the wearer

Instructor commentary at Different Dive describes using a paired alert device in practice. DBA separates device receipt from the wearer's explicit acknowledgement and leaves cuff transmission and glove handling as human-factors questions.

<https://differentdive.com/week-26-i-test-the-buddy-watcher/>

## Positioning has physical constraints

Garmin's SubWave limitations document describes environmental and geometry constraints for its own positioning system. It is evidence that acoustic navigation needs quality handling, not a transfer of Garmin specifications or compatibility to DBA.

<https://www.garmin.com/en-US/legal/subwavedisclaimer/>

## Vertical rate is not decompression protection

Divers Alert Network discusses ascent rates within broader decompression considerations. DBA provides a proposed rate display without claiming a safe ascent limit, a decompression model or prevention of decompression sickness.

<https://dan.org/alert-diver/article/ascent-rates/>



# A pair of bands. A shared reference.

The proposition and its present evidence.

The navigation concept adds a boat-associated acoustic positioning reference to the paired bands and inductive dock. A dive operator would deploy that reference; each diver would wear a band. This adds equipment and setup, while making buddy bearing and boat-relative position possible in the proposed system.

The value proposition is a private buddy call with readable spatial context. Demand, operator workflow, reference coverage, runtime and the complete system's price remain unestablished. The presentation defines a product direction for discussion, not a finished underwater navigation instrument.

## System economics require a new estimate

The navigation display, sensor additions and acoustic reference change the scope. Rev F has no supplier quote, complete cost estimate, retail target or established margin. Its reference-deployment workflow and customer demand remain open.

## Historical Rev A basis

The earlier display-free concept used about \$86 per pair at 5,000 pairs,  $\pm 30\%$ , and a \$179-199 retail target per pair. Those desk estimates exclude tooling, engineering, freight and duty. They do not price this navigation system.

## Concept documentation

The package supports discussion with product, acoustic, electronic and mechanical engineers. The source produces nominal CAD and presentation artifacts. No physical operation, pressure sealing, sensor accuracy, wearability or manufacturing readiness is established.

# Files for presentation and review

Current release: Presentation Rev F / CAD Rev F.

## **DBA\_Interactive.html**

Primary presentation: four simulated navigation views, actual 3D strap variants, dimensions, engineering register and downloads. The side depth gauge follows page scrolling; it is a presentation effect.

## **DBA\_Product\_Overview\_RevF.pptx**

Editable 12-slide overview with presenter notes. slides.html provides an optional browser slide viewer.

## **DBA\_Design\_Package\_RevF.pdf**

This concept and engineering brief, including proposed system architecture, controls, failure behavior, fit and source references.

## **DBA\_Annotated\_Views\_RevF.pdf**

Current top, front, internal section, sensor layout and three-size fit views. drawings/DBA-001\_drawing\_sheet.pdf provides the coordinated engineering sheet.

## **cad/ and cad/fit-variants/**

Regular assembly and part STEP/STL files, with Compact and Cuff variants. design-data.json records parameters and envelopes. Reference array architecture has no physical CAD design.

## **dba\_cad.py, render.py and drawing\_sheet.py**

Change named parameters in the source, regenerate CAD, then regenerate renders, drawings and embedded meshes. Do not hand-edit STEP or STL files.

## **presentation/ and README.txt**

Presentation source, requirements, screen logic and release-generation instructions. All browser mesh variants retain base64 int16 encoding.

## **legacy/**

Original Rev A package and earlier briefs are historical references. Current downloads and active presentation use Rev F.

## **Prepared for Rafael**

Presentation strategy, design & production by The AI Doc